ASSIGNMENT-13.2

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BATCH-04

TASK-01

**Task Description -1 (Refactoring – Eliminating Repetitive Logic)**

- Task: Use AI-assisted coding techniques to restructure a Python program that performs repeated calculations, transforming it into a reusable and maintainable design.

**Instructions:**

- Ask AI to locate repeated computations within the program.
- Encapsulate repeated logic into reusable functions.
- Verify that program output remains unchanged after refactoring.
- Add proper docstrings to all defined functions.

**Sample Code:**

```
length = 8
width = 4
print("Area:", length * width)
print("Perimeter:", 2 * (length + width))

length = 12
width = 6
print("Area:", length * width)
print("Perimeter:", 2 * (length + width))
```

**Expected Output:**

A refactored Python script that uses a single reusable function to compute area and perimeter without repeating code.

PROMPT:

Refactor the Python code by identifying repeated calculations and creating a reusable method to compute area and perimeter without changing the output.

ORIGINAL CODE:

```
lab_14_5.py > ...
length = 8
width = 4
print("Area:", length * width)
print("Perimeter:", 2 * (length + width))

length = 12
width = 6
print("Area:", length * width)
print("Perimeter:", 2 * (length + width))
```

OUTPUT:

```
Area: 32
Perimeter: 24
Area: 72
Perimeter: 36
```

REFACTORED CODE:

```
lab_14_5.py > ...
def rectangle_details(length, width):
    """
    Calculate and print the area and perimeter of a rectangle.

    Parameters:
    length (int/float): Length of the rectangle
    width (int/float): Width of the rectangle
    """

    area = length * width
    perimeter = 2 * (length + width)

    print("Area:", area)
    print("Perimeter:", perimeter)


# First rectangle
rectangle_details(8, 4)

# Second rectangle
rectangle_details(12, 6)
```

OUTPUT:

```
Area: 32
Perimeter: 24
Area: 72
Perimeter: 36
```

JUSTIFICATION:

**Eliminating Repetitive Logic**

Refactoring removes duplicate calculations by placing them inside a reusable method. This improves code maintainability and makes it easier to reuse the same logic for different inputs.

TASK2:

**Task Description -2 (Refactoring – Modularizing Inline Calculations)**

- Task: Use AI to refactor a Python script by extracting repeated inline calculations into reusable and well-documented functions.

**Instructions:**

- Identify repeated calculation patterns in the code.
- Convert them into reusable functions.
- Ensure readability and proper documentation.

**Sample Code:**

```
amount = 1000
discount = amount * 0.10
final_amount = amount - discount
print("Final Amount:", final_amount)
```

```
amount = 2000
discount = amount * 0.10
final_amount = amount - discount
print("Final Amount:", final_amount)
```

**Expected Output:**

A modular Python program that uses a reusable function to compute final amounts for different inputs.

PROMPT:

Extract the repeated discount calculation into a reusable Python method that computes the final amount after a 10% discount.

ORIGINAL CODE:

```
lab_14_5.py > ...
amount = 1000
discount = amount * 0.10
final_amount = amount - discount
print("Final Amount:", final_amount)

amount = 2000
discount = amount * 0.10
final_amount = amount - discount
print("Final Amount:", final_amount)
```

OUTPUT:

```
Final Amount: 900.0
Final Amount: 1800.0
```

REFACTORED CODE:

```
lab_14_5.py > calculate_final_amount
def calculate_final_amount(amount):
    """
    Calculate the final amount after applying a 10% discount.
    Parameters:
    amount (int/float): Original amount
    Returns:
    float: Final amount after discount
    """
    discount = amount * 0.10
    final_amount = amount - discount
    return final_amount
print("Final Amount:", calculate_final_amount(1000))
print("Final Amount:", calculate_final_amount(2000))
```

OUTPUT:

```
Final Amount: 900.0
Final Amount: 1800.0
```

JUSTIFICATION:

**Modularizing Inline Calculations**

Creating a separate method for the discount calculation reduces code repetition and improves modularity. It also makes the program easier to update if the discount logic changes.

TASK3:

**Task Description -3 (Refactoring – Improving Loop and Conditional Performance)**

- Task: Apply AI-based suggestions to enhance the performance of Python code that relies on nested loops and repeated conditional checks.

**Instructions:**

- Use AI to recommend more efficient data structures or logic simplifications.
- Replace inefficient constructs while preserving program behavior.
- Compare execution efficiency before and after refactoring.

**Sample Code:**

```
students = ["Anil", "Bhavya", "Charan", "Divya"]
check_list = ["Charan", "Esha", "Bhavya"]

for name in check_list:
    exists = False
    for student in students:
        if name == student:
            exists = True
    if exists:
        print(name, "found")
    else:
print(name, "not found")
```

**Expected Output:**

An optimized Python program that performs faster lookups using efficient logic, along with execution performance comparison.

PROMPT:

Optimize the Python code that uses nested loops for searching by using a more efficient data structure while keeping the same output.

ORIGINAL CODE:

lab_14_5.py > ...

```python
students = ["Anil", "Bhavya", "Charan", "Divya"]
check_list = ["Charan", "Esha", "Bhavya"]

for name in check_list:
    exists = False
    for student in students:
        if name == student:
            exists = True

    if exists:
        print(name, "found")
    else:
        print(name, "not found")
```

OUTPUT:

```
Charan found
Esha not found
Bhavya found
```

REFACTORED CODE:

lab_14_5.py > ...

```python
students = ["Anil", "Bhavya", "Charan", "Divya"]
check_list = ["Charan", "Esha", "Bhavya"]

student_set = set(students)  # faster lookup

for name in check_list:
    if name in student_set:
        print(name, "found")
    else:
        print(name, "not found")
```

OUTPUT:

```
Charan found
Esha not found
Bhavya found
```

JUSTIFICATION:

**Improving Loop and Conditional Performance**

Using a **HashSet** instead of nested loops improves lookup efficiency. This reduces time complexity and enhances the overall performance of the program.

TASK4:

**Task Description -4 (Refactoring – Replacing Hardcoded Values with Constants)**

- Task: Utilize AI assistance to improve maintainability by replacing hardcoded numerical values with named constants.

**Instructions:**

- Detect hardcoded numeric values used directly in expressions.
- Define descriptive constants at the beginning of the program.
- Update calculations to use these constants without changing results.

**Sample Code:**

```
print("Simple Interest:", (1000 * 2 * 5) / 100)
print("Simple Interest:", (2000 * 2 * 5) / 100)
```

**Expected Output:**

A revised Python program using constants such as RATE and TIME for clearer and maintainable calculations..

PROMPT:

Identify hardcoded numerical values used in the Python program and replace them with descriptive constants. Update the program so that the calculations remain the same but the code becomes more readable and maintainable.

ORIGINAL CODE:

lab_14_5.py

```
print("Simple Interest:", (1000 * 2 * 5) / 100)
print("Simple Interest:", (2000 * 2 * 5) / 100)
```

CODE:

```
Simple Interest: 100.0
Simple Interest: 200.0
```

REFACTORED CODE:

```
lab_14_5.py > ...
RATE = 2
TIME = 5
DIVISOR = 100

def calculate_interest(principal):
    """Calculate simple interest."""
    return (principal * RATE * TIME) / DIVISOR

print("Simple Interest:", calculate_interest(1000))
print("Simple Interest:", calculate_interest(2000))
```

CODE:

```
Simple Interest: 100.0
Simple Interest: 200.0
```

JUSTIFICATION:

**Replacing Hardcoded Values with Constants**

Replacing hardcoded values with named constants improves readability and maintainability. It also allows easy modification of values without changing multiple parts of the code.

TASK5:

**Task Description -5 (Re factoring – Enhancing Variable Naming and Code Clarity)**

- Task: Apply AI tools to improve code readability by renaming unclear variables and adding meaningful comments..

**Instructions:**

- Replace non-descriptive variable names with meaningful identifiers.
- Add inline comments where the logic may be unclear.
- Ensure program output remains the same.

**Sample Code:**

```
x = 50
y = 30
z = x + y
print(z)
```

**Expected Output:**

A Python program with clear variable names and explanatory comments that improve readability without altering functionality.

PROMPT:

Improve the Python code readability by renaming unclear variables and adding meaningful comments without changing the program functionality.

ORIGINAL CODE:

lab_14_5.py > ...

```
x = 50
y = 30
z = x + y
print(z)
```

OUTPUT:

```
80
```

REFACTORED CODE:

```
lab_14_5.py > ...
# Store two numbers
first_number = 50
second_number = 30

# Calculate their sum
total_sum = first_number + second_number

# Print the result
print(total_sum)
```

OUTPUT:

```
80
```

JUSTIFICATION:

**Enhancing Variable Naming and Code Clarity**

Using meaningful variable names and adding comments improves code readability. This helps developers understand the program logic quickly and reduces confusion during maintenance.